

Deva Anand

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Education

University of Massachusetts Amherst

Master of Science in Computer Science

September 2025 – May 2027 (Expected)

Amherst, MA

Coursework: Systems for Data Science, Performance Engineering, Optimization in CS, Advanced Machine Learning, Advanced NLP

Vels Institute of Science, Technology and Advanced Studies

Bachelor of Engineering in Computer Science and Engineering

August 2019 – May 2023

Chennai, India

Skills

Languages

TypeScript, JavaScript, Python, Java, C++, SQL, Bash, HTML/CSS

Backend & APIs

Node.js, Express, Fastify, FastAPI, REST APIs, authentication, rate limiting

Web & Databases

React, Next.js, Electron, MongoDB, PostgreSQL, Redis, SQLite

AI & GenAI

LLM agents, RAG, retrieval, structured extraction with verification & provenance, OpenAI / Groq APIs, PyTorch, Hugging-Face Transformers, Claude Code / Copilot / Cursor

Cloud & DevOps

AWS, GCP, Azure, Docker, Git, GitHub Actions, CI/CD, Linux

Experience

Wysa

Backend Engineer

July 2023 – July 2025

Bengaluru, India

- Engineered **multi-tenant NHS clinical backend** for **8+ UK clinical clients** (Node.js, MongoDB) powering self-referral and triage workflows that process **10,000+ monthly submissions**, integrated with an external clinical platform (Mayden's iaptus) via REST APIs, contributing to **\$340K+** in revenue.
- Owned backend security for sensitive clinical data: hardened **authentication, rate limiting, and audit logging**, added integration tests for regression protection, monitored production logs during releases and client integrations, and remediated **20+ Vulnerability Assessment and Penetration Testing (VAPT) findings**.
- Architected a **full-stack data annotation tool (React, Node.js, MongoDB)** used daily by the AI team to label data for production models, replacing spreadsheet workflows with secure role-based REST APIs; cut manual operations by **100+ hours/month** and improved labeling throughput by **60%**.

Cario Growth Services Pvt. Ltd.

Backend Developer Intern

November 2022 – May 2023

Chennai, India

- Designed and shipped **20+ RESTful APIs** in Node.js and Fastify, backed by **PostgreSQL**, with input validation, clear contracts, and backward-compatible changes used in production by the frontend team.
- Migrated core codebase from **JavaScript to TypeScript** across multiple repositories and integrated **LLM-powered context-aware replies** into a chat product, iterating on prompt design and response-quality heuristics for a real customer-facing experience.

Projects

AgenticSearch - Grounded, Provenance-First Entity Discovery

github.com/Deva-1903/ciir_agentic_search

Python, FastAPI, Brave Search, OpenAI / Groq, SQLite

- Built a multi-stage pipeline that integrates external sources via web search and retrieval, then produces **structured AI outputs** through deterministic parsers with **LLM fallback**, output verification, and **cell-level provenance**, grounded AI rather than a single opaque LLM call.
- Hardened the FastAPI service with **150+ automated tests**, provider routing and fallback across LLM backends, pipeline instrumentation, and a reviewer-facing trust UI for inspecting extraction quality and ranking transparency.

Sonare - Offline Sign ↔ Speech Cross-Platform Desktop App

github.com/Deva-1903/Qualcomm-Sep25-Team-Sonare

React, Electron, FastAPI, MediaPipe, whisper.cpp | Qualcomm Edge AI Hackathon 2025

- Built a **full-stack desktop application** integrating camera, microphone, UI, and backend services into an on-device pipeline using MediaPipe hand tracking, gesture classification, whisper.cpp ASR, and local Text-To-Speech, designed and shipped end-to-end under hackathon time pressure.
- Engineered a **FastAPI backend** and **Electron/React frontend** with async APIs and queues that keep the UI responsive while streaming local video and audio with no external network dependencies.

cf-ai-research-scout - Evidence-Retrieval AI Research Assistant

github.com/Deva-1903/cf_ai_research_scout

TypeScript, Cloudflare Workers AI, WebSockets, SQL

- Built an AI research assistant on serverless cloud functions: the agent fetches external web sources, generates multi-step research digests, and streams responses over **WebSocket** in real time.
- Designed durable workflow boundaries so multi-step research tasks survive process restarts and tool failures; agent state persists server-side without an external database.

Publication & Extracurriculars

- Publication:** "Alzheimer's Disease Classification using Transfer Learning," **IEEE CONIT 2023**, applied deep transfer learning on neuroimaging data for clinical medical-image classification.
- Selected for **GirlScript Summer of Code (GSSoC) 2023** and contributed to multiple open-source projects (Linkfree, Freehit, ProjectsHut).
- Tutored **30+** underprivileged students in programming at Sayur, non-profit in South Tamil Nadu.